

# Forecasting with ARIMA models

FISH 550 – Applied Time Series Analysis   [Download Rmd pdf](#)

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# Forecasting with an ARIMA model

The basic idea of forecasting with an ARIMA model to estimate the parameters and forecast forward.

For example, let's say we want to forecast with a ARIMA(2,1,0) model with drift:

$$z_t = \mu + \beta_1 z_{t-1} + \beta_2 z_{t-2} + e_t$$

where  $z_t = x_t - x_{t-1}$ , the first difference.

`Arima()` would write this model:

$$(z_t - m) = \beta_1(z_{t-1} - m) + \beta_2(z_{t-2} - m) + e_t$$

The relationship between  $\mu$  and  $m$  is  $\mu = m(1 - \beta_1 - \beta_2)$ .

Let's estimate the  $\beta$ 's for this model from the anchovy data.

```
fit <- forecast::Arima(anchovyts, order=c(2,1,0), include.drift=TRUE)
coef(fit)
```

```
##           ar1           ar2           drift
## -0.53850433 -0.44732522  0.05367062
```

```
mu <- coef(fit)[3]*(1-coef(fit)[1]-coef(fit)[2])
mu
```

```
##           drift
## 0.1065807
```

So we will forecast with this model:

$$z_t = 0.1065807 - 0.53850433z_{t-1} - 0.44732522z_{t-2} + e_t$$

To get our forecast for 1990, we do this

$$(x_{90} - x_{89}) = 0.106 - 0.538(x_{89} - x_{88}) - 0.447(x_{88} - x_{87})$$

Thus

$$x_{90} = x_{89} + 0.106 - 0.538(x_{89} - x_{88}) - 0.447(x_{88} - x_{87})$$

Here is R code to do that:

```
anchovyts[26] + mu + coef(fit)[1] * (anchovyts[26] - anchovyts[25]) +  
  coef(fit)[2] * (anchovyts[25] - anchovyts[24])
```

```
##      drift  
## 9.962083
```

## Forecasting with forecast()

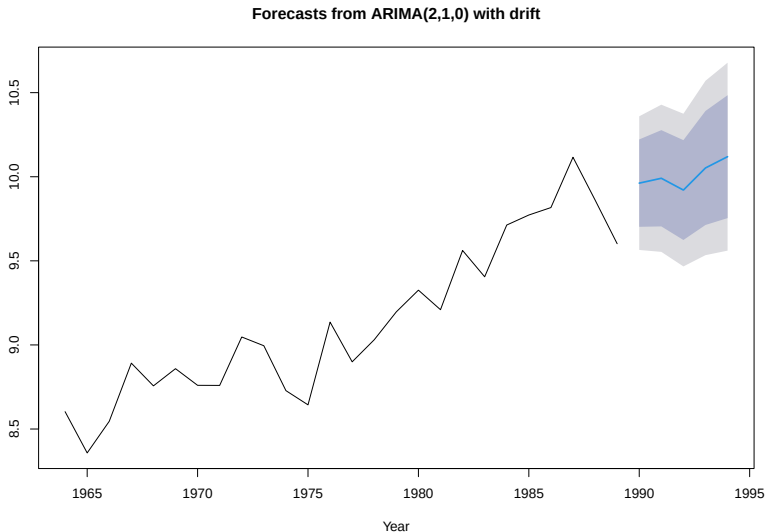
`forecast(fit, h=h)` automates the forecast calculations for us and computes the upper and lower prediction intervals. Prediction intervals include uncertainty in parameter estimates plus the process error uncertainty.

```
fr <- forecast::forecast(fit, h=5)
fr
```

| ##      | Point Forecast | Lo 80    | Hi 80    | Lo 95    | Hi 95    |
|---------|----------------|----------|----------|----------|----------|
| ## 1990 | 9.962083       | 9.702309 | 10.22186 | 9.564793 | 10.35937 |
| ## 1991 | 9.990922       | 9.704819 | 10.27703 | 9.553365 | 10.42848 |
| ## 1992 | 9.920798       | 9.623984 | 10.21761 | 9.466861 | 10.37473 |
| ## 1993 | 10.052240      | 9.713327 | 10.39115 | 9.533917 | 10.57056 |
| ## 1994 | 10.119407      | 9.754101 | 10.48471 | 9.560719 | 10.67809 |

# Plotting our forecasts

```
plot(fr, xlab="Year")
```



## Missing values

Missing values are allowed for `forecast::Arima()`. We can produce forecasts with the same code.

```
anchovy.miss <- anchovyts
anchovy.miss[10:11] <- NA
anchovy.miss[20:21] <- NA
fit <- forecast::Arima(anchovy.miss, order=c(2,1,0), include.mean=TRUE)
fr <- forecast::forecast(fit, h=5)
fr
```

|         | Point Forecast | Lo 80    | Hi 80    | Lo 95    | Hi 95    |
|---------|----------------|----------|----------|----------|----------|
| ## 1990 | 9.938269       | 9.664479 | 10.21206 | 9.519543 | 10.35700 |
| ## 1991 | 10.014686      | 9.700961 | 10.32841 | 9.534885 | 10.49449 |
| ## 1992 | 9.924208       | 9.601147 | 10.24727 | 9.430129 | 10.41829 |
| ## 1993 | 10.029988      | 9.666069 | 10.39391 | 9.473421 | 10.58656 |
| ## 1994 | 10.128066      | 9.729066 | 10.52707 | 9.517848 | 10.73828 |

```
plot(fr)
```

Forecasts from ARIMA(2,1,0) with drift



## Using auto.arima()

We can let forecast to select the ARIMA model:

```
anchovy.miss <- anchovyts  
anchovy.miss[10:11] <- NA  
anchovy.miss[20:21] <- NA  
fit <- forecast::auto.arima(anchovy.miss)  
fit
```

```
## Series: anchovy.miss  
## ARIMA(0,1,1) with drift  
##  
## Coefficients:  
##          ma1    drift  
##      -0.7240  0.0548  
## s.e.    0.2283  0.0125  
##  
## sigma^2 = 0.04355:  log likelihood = 3.52  
## AIC=-1.03    AICc=0.11    BIC=2.63
```

```
fr <- forecast::forecast(fit, h=5)  
plot(fr)
```

